

DistriNVS: Distributional Geometry Transport and Visibility-Constrained Completion for Cross-Endoscope Novel-View Synthesis

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Abstract—Cross-endoscope novel-view synthesis must both transport tissue observed by the source camera and infer tissue revealed only from the target viewpoint. These two regimes are confounded when sparse or erroneous depth is projected as deterministic points: sampling gaps, depth failures, surface collisions, and genuine disocclusions all become visually identical holes. We present DistriNVS, a reliability-aware framework that models every source depth as a distribution and propagates it through calibrated geometry. Its distributional surface splatting renderer produces an anisotropic target-plane footprint together with support, depth variance, and collision entropy. These statistics drive reliability-constrained transport-synthesis fusion: a bounded residual expert refines transported appearance, whereas an uncertainty-conditioned Fourier network predicts unsupported or geometrically unreliable content. Their reliability-weighted RGB-D mixture and an explicit risk head are trained with target-depth and backward-reprojection constraints. The prescribed protocol trains on a fixed set of 13 iMED scenes and evaluates without EndoVis adaptation on 275 frozen pairs from held-out EndoVis Datasets 8 and 9, each satisfying a strict raw-overlap threshold of 0.70.

I. INTRODUCTION

Robot-assisted minimally invasive surgery depends on endoscopic vision for scene perception and instrument guidance [1], [2]. Cross-endoscope novel-view synthesis can broaden this visual context by predicting a target-endoscope view from synchronized appearance and geometry acquired by a spatially displaced source endoscope [3]. Unlike left-to-right synthesis within one stereo endoscope, the target may observe anatomy absent from both source cameras. The task therefore combines two epistemically distinct regimes: co-visible anatomy has measured appearance that should be transported through calibrated geometry, whereas target-only anatomy lacks source evidence and must be synthesized. A reliable system should retain source evidence, identify inferred content, and thereby support wider-field robotic perception, retrospective visualization, and surgical simulation [4]. Such uncertainty-aware cross-view completion may also support visual-state construction for surgical robot world models [5].

Achieving this separation is difficult because endoscopic depth and calibration are unreliable around specularities, weak texture, depth discontinuities, instruments, and deforming tissue [6]. Conventional forward warping represents each valid source pixel by one 3-D point; discretized projection leaves sampling gaps, while multiple points may compete for the same target pixel. A binary validity mask then conflates missing depth, rasterization holes, depth-ordering conflicts,

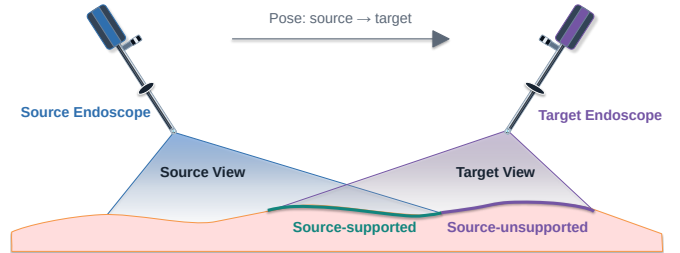


Fig. 1. Cross-endoscope novel-view synthesis under partial field-of-view overlap. Given the source-to-target pose, co-visible anatomy is source-supported and can be transported geometrically (■ teal), whereas target-only anatomy lacks source evidence and needs to be synthesized (■ purple).

and genuine cross-endoscope disocclusions. Treating these cases uniformly either degrades observed structure or permits hallucination where geometric evidence remains trustworthy.

Existing methods address complementary parts of this problem but do not expose the required evidence boundary. Differentiable point rendering and softmax splatting improve feature transport and collision handling [7], [8], while reference-guided inpainting combines depth-based alignment with learned completion [9]. Large-mask inpainting networks provide strong global image priors [10], [11], and endoscopic video completion adds depth, temporal, or stereo cues [12]–[14]. Endoscopic neural rendering has likewise advanced sparse-view reconstruction and uncertainty modeling [15], [16]. However, deterministic projection commonly assigns equal reliability to all valid depths, while image-space uncertainty does not explicitly describe how source geometry supports a spatially offset target view. Neither approach propagates per-pixel depth uncertainty through the cross-endoscope transform and uses the resulting visibility statistics to constrain where appearance may be synthesized.

Our central premise is to *transport reliable geometric evidence and synthesize only unsupported or geometrically unreliable regions*. We introduce DistriNVS, a reliability-aware framework for direct cross-endoscope novel-view synthesis. Its distributional surface splatting (DSS) represents each source depth as a distribution whose bounded mean and scale, surface normal, and confidence are predicted by a depth reliability encoder. DSS projects deterministic sigma samples through the calibrated cross-endoscope transform and aggregates source features using Jacobian- and normal-aware footprints with soft nearest-surface visibility, yielding target support, depth variance, and collision entropy. These statistics constrain a soft mixture between bounded residual

transport for co-visible regions and uncertainty-conditioned Fourier completion for unsupported or ambiguous regions. Target-depth and risk prediction plus backward RGB-D reprojection constrain this mixture during training.

The paper makes three contributions:

- We formulate cross-endoscope novel-view synthesis as a reliability-aware separation between transport of anatomy observed by the source endoscope and constrained inference of content revealed only by the target endoscope.
- We introduce DSS, a differentiable renderer that propagates depth distributions through calibrated cross-endoscope geometry and exposes target support, depth variance, and collision entropy beyond binary validity.
- We design a two-expert decoder that softly mixes bounded residual transport with uncertainty-conditioned Fourier completion under a renderer-derived reliability constraint.

II. RELATED WORK

We briefly review image completion, geometry-guided view transport, and endoscopic neural rendering, with particular attention to how existing approaches represent missing content, exploit calibrated geometry, and quantify uncertainty.

A. Image and Endoscopic Video Inpainting

Generic image inpainting combines mask-aware feature extraction with increasingly broad appearance priors. Partial and gated convolutions prevent invalid activations from contaminating local features [17], [18], whereas LaMa and UFFC use Fourier operations for global context and MAT restricts attention to valid tokens [10], [11], [19]. Recent diffusion models such as BrushNet and TurboFill incorporate masked-image features into pretrained generators or adapt few-step sampling for efficient completion [20], [21]. Despite strong appearance priors, these methods define holes in a fixed image coordinate system and do not describe geometric provenance. Endoscopic completion adds domain cues: temporal learning removes specularities, DAEVI combines spatio-temporal RGB and depth, and SSIFNet fuses flow and stereo-temporal information for instrument occlusions [12]–[14]. Such methods exploit neighboring frames or views but still restore defects defined in the original camera coordinates. Cross-endoscope synthesis instead creates holes through viewpoint transfer: target-only anatomy is unobserved, while rasterization gaps or depth conflicts may retain source support. DistriNVS therefore uses calibrated projection statistics to distinguish true disocclusions from rasterization gaps and depth conflicts.

B. Geometry-Guided Transport and Completion

Geometry-guided view synthesis preserves correspondence by transporting source evidence before generating missing content. SynSin predicts and projects a feature point cloud, softmax splatting resolves many-to-one mappings with learned importance, and GeoFill refines depth

and pose before reprojection and inpainting [7]–[9]. Recent feed-forward Gaussian renderers use stronger explicit representations: MVSplat localizes Gaussians with plane-sweep cost volumes, whereas DepthSplat combines multi-view correspondence with monocular depth priors [22], [23]. They reduce per-scene optimization but infer a radiance representation from multiple calibrated observations, without an explicit per-pixel boundary between measured and synthesized appearance. Generative methods target wider gaps. LeftRefill frames reference-conditioned synthesis as paired-canvas completion, MultiDiff conditions video diffusion on depth-warped reference views, and point-guided latent diffusion uses stereo-derived clouds for laparoscopic view synthesis [24]–[26]. These models can generate plausible structures in large disocclusions, yet plausibility alone does not reveal which pixels remain geometrically supported. DistriNVS instead transports a calibrated source observation with distributional splatting, derives support, variance, and collision statistics during rasterization, and uses them to constrain soft fusion between transported and synthesized content.

C. Endoscopic Novel-View Synthesis and Uncertainty

Endoscopic neural rendering must handle non-rigid tissue, view-dependent illumination, tool occlusions, and sparse camera coverage. EndoNeRF represents deformation with a neural radiance field, while EndoSurf combines dynamic modeling with a neural surface [27], [28]. EndoSparse introduces geometric and appearance priors into Gaussian splatting, and Endo-GSMT uses depth, dense-displacement, and motion priors for dynamic reconstruction [15], [29]. GastroNVS further shows that large viewpoint changes in real gastroscopy amplify geometric artifacts and inter-view illumination mismatch [30]. These systems build persistent scene representations from multiple frames and principally address reconstruction fidelity within a captured scene. Uncertainty-aware approaches address sparse evidence from another direction: UC-NeRF conditions a radiance field on multi-view uncertainty, whereas ExtraGS uses uncertainty to sample virtual cameras and weight diffusion-refined pseudo-observations for extrapolation [4], [16]. Thus, neither uncertainty nor generative priors are new to endoscopic rendering. The distinction is operational: DSS propagates a distribution for each input depth through the calibrated cross-endoscope transform and derives target support, depth variance, and collision entropy from the same rasterization that transports appearance. DistriNVS uses these quantities to route completion during unseen-sequence inference without target-scene optimization.

III. METHOD

A. Problem Formulation and Reliability-Aware Geometry

After resizing, both endoscope views have spatial size $H \times W$; we omit the batch dimension throughout. DistriNVS receives source RGB $\mathbf{I}_s \in [0, 1]^{3 \times H \times W}$, sparse metric depth $\mathbf{D}_s \in \mathbb{R}_+^{1 \times H \times W}$, and validity mask $\mathbf{M}_s \in \{0, 1\}^{1 \times H \times W}$. Calibration comprises intrinsics $\mathbf{K}_s, \mathbf{K}_t \in \mathbb{R}^{3 \times 3}$ and a rigid

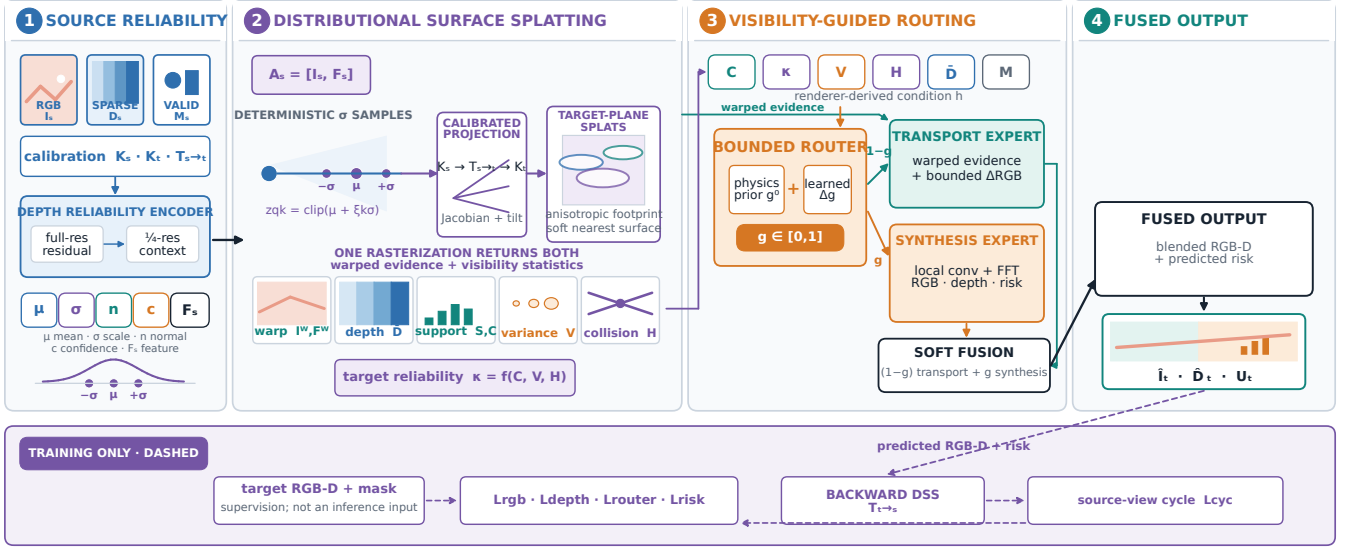


Fig. 2. DistriNVS framework. A reliability encoder predicts source geometry and features. DSS transports deterministic depth hypotheses through the calibrated cross-endoscope transform and jointly produces warped evidence and target-plane visibility statistics. These statistics condition a reliability-constrained soft mixture of a residual transport expert and a Fourier synthesis expert. Solid arrows denote inference; the dashed lower lane is used only during training.

transform $\mathbf{T}_{s \rightarrow t} \in \text{SE}(3) \subset \mathbb{R}^{4 \times 4}$. The model predicts target RGB $\hat{\mathbf{I}}_t \in [0, 1]^{3 \times H \times W}$, depth $\hat{\mathbf{D}}_t \in \mathbb{R}_+^{1 \times H \times W}$, and risk $\mathbf{U}_t \in [0, 1]^{1 \times H \times W}$. Target RGB-D and the target instrument mask are used only for training or evaluation.

Figure 2 summarizes the inference pipeline. A reliability encoder estimates source geometry and uncertainty. Distributional surface splatting (DSS) then transports source evidence and measures target-view support and ambiguity. These measurements route complementary transport and synthesis experts, whose RGB-D predictions are combined by a reliability-constrained soft gate. The backward photometric cycle and risk objective are used only during training.

We first construct the source-side geometric representation on which DSS operates. Rather than treating measured and completed depths as equally reliable, the reliability encoder processes source RGB, median-normalized depth, and its validity mask. It produces a feature map $\mathbf{F}_s \in \mathbb{R}^{d_f \times H \times W}$ and scalar fields $\mu, \sigma, c \in \mathbb{R}^{1 \times H \times W}$. They represent depth mean and scale in millimetres, and normalized confidence. It also predicts a unit orientation field $\mathbf{n} \in \mathbb{R}^{3 \times H \times W}$. The normalization scale s_D is a positive scalar in millimetres, chosen as the median valid source depth, and missing inputs are set to zero. During training, $\mathbf{M}_s$ is the post-dropout input mask, whereas $\mathbf{M}_s^{\text{gt}}$ has the same dimensions and retains the pre-dropout supervision.

Measured depths receive a bounded residual correction; missing depths use a positive completion branch. Both obey the physical range, but disjoint confidence intervals keep inferred geometry less authoritative than measurements. The optical-axis-stabilized orientation proxy is learned only through downstream objectives. It serves as a footprint cue rather than a metrically supervised normal. The resulting $(\mu, \sigma, c, \mathbf{n}, \mathbf{F}_s)$ parameterize the depth hypotheses, footprints,

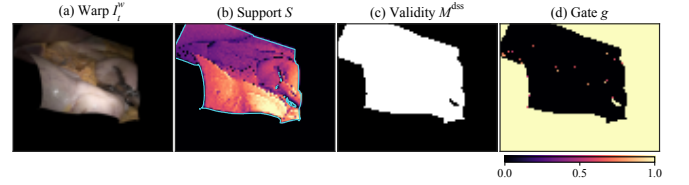


Fig. 3. DSS evidence and reliability-guided expert routing for a held-out iMED validation target frame with substantial source-view disocclusion. (a) The completed-geometry DSS warp $\mathbf{I}_t^w$; unsupported pixels are black. (b) Unnormalized support S , capped at the framewise q_{vis} percentile only for display; the cyan contour marks $S = S_{\min}$. (c) The exact render-validity mask M^{dss} , computed by thresholding uncapped support. (d) The effective synthesis gate g from Eq. (6). It is zero on reliable support, selecting the bounded transport expert, and assigns progressively greater weight to the synthesis expert as reliability decreases. The gate uses the full reliability condition rather than the complement of M^{dss} .

and transported attributes used by DSS. Fixed ranges and clipping operations appear in Sec. IV-B.

B. Distributional Surface Splatting

Given these source-side estimates, DSS concatenates source RGB and features as $\mathbf{A}_s = [\mathbf{I}_s, \mathbf{F}_s]$ and transports it to a target tensor $\mathbf{A}_t^w$. Both tensors contain $3 + d_f$ channels over the $H \times W$ grid. DSS simultaneously predicts expected depth $\hat{\mathbf{D}}_t \in \mathbb{R}_+^{1 \times H \times W}$. Support S , coverage C , depth variance V , collision entropy $\mathcal{H}$, confidence κ , and render validity M^{dss} are $1 \times H \times W$ target-grid maps. Support and variance are non-negative; coverage, collision entropy, and confidence lie in $[0, 1]$; validity is binary. All arise from one rasterization of a compact depth distribution at each source pixel. Expected depth is in millimetres and variance in square millimetres; the remaining evidence maps are dimensionless.

Depth hypotheses. For each source pixel q , depth hypothesis z_{qk} corresponds to a fixed standardized offset $\xi_k \in \Xi$ and

quadrature weight a_k . The set and weights are specified in Sec. IV-B. The calibrated pinhole operator projects each hypothesis to a target coordinate $\mathbf{x}_{qk} \in \mathbb{R}^2$ and target depth $z_{qk}^t \in \mathbb{R}_+$. It comprises source back-projection, rigid transformation, and target projection [7], [8].

Uncertainty-aware visibility. Depth uncertainty is converted into a hypothesis-dependent, axis-aligned target footprint. The projection derivative J_{qk}^α measures the sensitivity of image coordinate $\alpha \in \{u, v\}$ to source depth. It has units of pixels per millimetre. The pixel-space footprint is

$$r_{qk}^\alpha = b + |J_{qk}^\alpha| \sigma_q + b_n(1 - |n_q^z|). \quad (1)$$

The radii r_{qk}^α , b , and b_n are measured in pixels, while n_q^z is the optical-axis component of the orientation proxy. For target pixel p , ρ_{qkp} denotes its elliptical distance from the projected hypothesis, normalized by both radii. The corresponding contribution is

$$w_{qk \rightarrow p} = c_q a_k \exp \left[-\frac{\rho_{qkp}^2}{2} - \frac{[z_{qk}^t - z_{\min}(p)]_+}{\tau_z} \right]. \quad (2)$$

Source confidence and quadrature mass scale this contribution. Spatial proximity and soft nearest-surface visibility form the exponential penalties. Here, $z_{\min}(p)$ is the detached nearest arrival, $[\cdot]_+$ is the positive part, and τ_z is the visibility temperature in millimetres. Nearest-depth selection and integer assignment are discrete; all remaining operations are differentiable.

Reliability statistics. Let Ω_p collect the retained source-hypothesis contributions to target pixel p . Their total weight defines raw support S_p . At render-valid pixels, DSS normalizes these weights over Ω_p . Their weighted moments yield the transported tensor, expected depth, and conditional depth variance. Coverage is a bounded saturation of raw support. The normalization constant and exact coverage transform are reported in Sec. IV-B.

DSS couples normalized weight entropy to depth dispersion. Let $H_N(p)$ denote the normalized Shannon entropy of the retained weights at pixel p . Collision ambiguity is

$$\mathcal{H}_p = H_N(p) [1 - \exp(-V_p/\tau_z^2)] \quad (3)$$

This coupling discounts entropy caused only by benign sub-pixel overlap. We derive target confidence κ_p by penalizing coverage with depth variance and collision ambiguity. A pixel is render-valid when its raw support reaches $S_{\min}$, as shown in Fig. 3(c). Invalid pixels retain support and coverage but zero the transported outputs and ambiguity maps. The continuous evidence therefore distinguishes weak support from conflicting depth order rather than reducing reliability to a binary mask.

C. Reliability-Guided Expert Fusion

Physics-guided routing. Both experts receive transported evidence and a condition tensor $\mathbf{h} \in \mathbb{R}^{6 \times H \times W}$. Its channels are coverage, confidence, normalized variance, collision entropy, normalized depth, and DSS validity. An analytic prior

$g^0 \in [0, 1]^{1 \times H \times W}$ favours synthesis as geometric reliability decreases and equals one outside DSS support:

$$\begin{aligned} \phi_p &= \alpha_\kappa(1 - \kappa_p) + \alpha_V \bar{V}_p + \alpha_H \mathcal{H}_p, \\ g_p^0 &= \begin{cases} \text{clip}_{[0,1]}(\phi_p), & M_p^{\text{dss}} = 1, \\ 1, & M_p^{\text{dss}} = 0. \end{cases} \end{aligned} \quad (4)$$

The provisional synthesis probability is

$$\tilde{g}_p = \text{sigmoid}[\ell_p^0 + \eta_g \tanh R_\theta(\mathbf{h})_p]. \quad (5)$$

Here, $\ell^0 = \text{sg}[\text{logit}(\text{clip}_{[\epsilon_g, 1-\epsilon_g]} g^0)]$, and the residual router R_θ outputs a scalar target-grid map. The factor η_g bounds its correction. A binary routing mask M^{rel} accepts DSS evidence only where the render is valid, $S_p \geq S_{\text{rel}}$, and $V_p \leq V_{\text{rel}}$. The effective synthesis gate is

$$g = (1 - M^{\text{rel}}) \odot \tilde{g}. \quad (6)$$

Routing therefore retains a geometric initialization, suppresses synthesis on reliable DSS support, and adapts continuously elsewhere. The prior coefficients and thresholds are reported in Sec. IV-B. Figure 3(d) visualizes the resulting gate. It is not the complement of M^{dss} because coverage, confidence, variance, collision entropy, normalized depth, and the learned router also affect its value. All three maps have shape $1 \times H \times W$.

Complementary experts. The shared expert input has $(9 + d_f) \times H \times W$ channels. The transport expert predicts a bounded RGB correction to the DSS warp, yielding $\mathbf{I}_t^{\text{trans}} \in [0, 1]^{3 \times H \times W}$. It has no depth head, so its depth remains the DSS expectation. The synthesis expert is an uncertainty-conditioned Fourier U-Net [31] that combines local convolution with real-FFT processing. Its outputs $(\mathbf{I}_t^{\text{syn}}, \mathbf{D}_t^{\text{syn}}, \mathbf{U}_t^{\text{syn}})$ have channel dimensions 3, 1, and 1 over the target grid. This local-global design follows Fourier inpainting networks [10], [11], but all weights are trained within DistriNVS. Risk is directly trained as an error-scale proxy on unreliable regions, not as a calibrated failure probability over the full image.

Reliability-constrained output. The transport branch retains source evidence while allowing bounded appearance correction; the synthesis branch supplies content where geometric evidence is weak. Their final predictions are

$$\hat{\mathbf{I}}_t = (1 - g) \odot \mathbf{I}_t^{\text{trans}} + g \odot \mathbf{I}_t^{\text{syn}}, \quad (7)$$

$$\hat{\mathbf{D}}_t = (1 - g) \odot \bar{D}_t + g \odot \mathbf{D}_t^{\text{syn}}, \quad \mathbf{U}_t = \mathbf{U}_t^{\text{syn}}. \quad (8)$$

The gate is broadcast across RGB channels. On M^{rel} , $g = 0$ and the output follows the bounded transport expert rather than an exact copy of the warp; elsewhere the two experts form a soft mixture. For regional supervision, we open the unreliable complement and define $M^{\text{reg}} = M^{\text{dss}} \wedge \neg \text{Open}_k(1 - M^{\text{rel}})$, where M^{dss} is render validity. This cleanup removes isolated unreliable islands from the regional partition; it affects only losses and diagnostics, never the predicted RGB-D or the gate in Eq. (6).

D. Learning Objectives

The training target $\mathbf{I}_t \in [0, 1]^{3 \times H \times W}$ is accompanied by a tissue-valid mask $M_{\text{tis}} \in \{0, 1\}^{1 \times H \times W}$, which excludes annotated instruments when available. The unreliable, reliable, and seam regions are $M_u = (1 - M^{\text{reg}}) \wedge M_{\text{tis}}$, $M_v = M^{\text{reg}} \wedge M_{\text{tis}}$, and $M_b = \text{Band}_w(1 - M^{\text{reg}}) \wedge M_{\text{tis}}$, where Band_w is a width- w dilation-erosion band. Their Charbonnier terms compare predicted and target RGB; $\mathcal{L}_{\nabla}$ additionally matches finite-difference image gradients on $M_u \cup M_b$. Training uses this cleaned regional partition, whereas regional image-quality and risk metrics use the raw DSS hole mask.

A masked Laplace negative log-likelihood $\mathcal{L}_{d\text{-nll}}$ supervises the source depth mean and scale on the pre-dropout mask M_s^{gt} . Available target depth $\mathbf{D}_t$ shares the dimensions of $\hat{\mathbf{D}}_t$ and supplies a masked relative-depth term $\mathcal{L}_{t\text{-depth}}$. The depth-guided cycle splats the predicted RGB-D back through the inverse calibrated transform. Detached risk controls its depth scale, whereas complementary risk controls its confidence. The cycle term $\mathcal{L}_{\text{cyc}}$ compares backward-rendered RGB with the source image where both are valid. We do not apply a depth-cycle loss.

The detached router target is $y_p = \mathbf{1}\{S_p < S_{\text{rel}} \vee V_p > V_{\text{rel}} \vee \mathcal{H}_p > \mathcal{H}_{\text{rel}}\}$, and $\mathcal{L}_{\text{router}}$ applies binary cross-entropy to the provisional logit in Eq. (5) over the full image. Render-invalid pixels necessarily satisfy the support test because $S_{\text{rel}} > S_{\text{min}}$. For a mask M , $\langle \cdot \rangle_M$ denotes the mean over its selected entries. The scalar e_p is the channel-mean absolute RGB error at pixel p . The positive scale s_p is an affine mapping of predicted risk between fixed bounds. The uncertainty objective is

$$\mathcal{L}_u = \langle e_p / s_p + \log s_p \rangle_{M_u}. \quad (9)$$

Exact risk bounds are given in Sec. IV-B. The target-depth term is omitted when target depth is unavailable. The complete objective is

$$\begin{aligned} \mathcal{L} = & \lambda_h \mathcal{L}_h + \lambda_v \mathcal{L}_v + \lambda_b \mathcal{L}_b + \lambda_{\nabla} \mathcal{L}_{\nabla} \\ & + \lambda_d (\mathcal{L}_{d\text{-nll}} + \mathcal{L}_{t\text{-depth}}) + \lambda_c \mathcal{L}_{\text{cyc}} \\ & + \lambda_r \mathcal{L}_{\text{router}} + \lambda_u \mathcal{L}_u. \end{aligned} \quad (10)$$

At inference, the predictor consumes only source RGB-D, camera calibration, and relative pose, and ends at the fused prediction in Eq. (8). Target RGB-D and instrument masks are used only for supervision or evaluation; the backward-cycle branch and deterministic image repair are disabled. The method assumes known zero-skew pinhole intrinsics, a rigid inter-endoscope transform, and an axis-aligned approximation to projected uncertainty; these assumptions delimit its present scope.

IV. EXPERIMENTS

A. Datasets and Zero-Shot Protocol

DistriNVS training uses the frozen list of 13 iMED Task 2 scene directories supplied in the project configuration [32]. Splits are made at scene level, not at frame level. The training loader uses a synchronized source RGB-depth and

target RGB pair at 640×512 resolution. Fifteen percent of valid source depth samples are removed online, with additional 9×9 missing islands, so that the reliability encoder observes both isolated and contiguous depth failures. EndoVis/SCARED datasets 8 and 9 are reserved for final zero-shot evaluation; neither their images nor metrics are used for model selection.

For this final zero-shot test, we determine inclusion using a renderer-only pass from the *raw* valid source depth. We retain a frozen set of 275 pairs for which $|M_{\text{raw}}^{\text{dss}}| / (H_I W_I) > 0.70$; learned depth completion, the target image, and the target tool mask cannot influence this decision. The same scene-frame identifiers are passed to every method, with no method-side or evaluator-side frame rejection. The tables report frame-wise means and sample standard deviations, both rounded to two decimal places.

B. Baselines, Metrics, and Implementation

Geometry baselines will include hard point projection, nearest-neighbour small-hole filling, deterministic soft splatting, and softmax splatting. Image completion comparisons include LaMa, a latent diffusion model (LDM) instantiated with the Stable Diffusion v1.5 Inpainting checkpoint, and an endoscopy-adapted LCM-LoRA system comprising the pretrained LCM-LoRA acceleration module and a training-split domain LoRA [10], [33], [34]. We additionally report an iMED-adapted Hunyuan-DiT [35] checkpoint. It is an in-domain diagnostic in Table I, but a cross-dataset zero-shot comparator in Table II, since no EndoVis sample is used for its adaptation. Directly relevant endoscopic baselines (DAEVI and SSIFNet) will be evaluated when code and an equivalent cross-view protocol are available; otherwise their non-comparable reported values will not be inserted into the main table. Every completion prediction is evaluated at the same spatial resolution on the same selected frames and targets, without frame rejection or evaluator-side image repair. On the iMED split, full metrics use common non-tool tissue support and H metrics intersect it with the canonical primary-run hole mask. The generic completion input mask covers 25.84% of an iMED image on average and is kept separate from this H mask. The target tool-mask directory is empty for the 199 iMED frames of *session_006_scene_7_tool_3*; those frames therefore use the full image as tissue support for every method. On EndoVis, where target tool masks are unavailable in the prepared benchmark, full metrics use all pixels and H metrics use the canonical raw-DSS hole mask without a tool-mask intersection.

For the Hunyuan-DiT comparator, we load the step-5,000 ControlNet and DiT LoRA and use a 100-step DDPM schedule with masked-latent strength 0.75 (75 denoising iterations), guidance scale 1.0 (no classifier-free guidance), zero mask feathering, and seed 42. Its binary input mask controls synthesis, and the Hunyuan-DiT comparator hard-composites the input warp over the complementary known region; this input mask is not used to define the common H metrics. Four GPUs process fixed disjoint shards in batches

of four. All reported scores are recomputed by the common evaluators used for Tables I and II, rather than the model’s built-in validation code.

The existing evaluator reports full-, raw-hole-, raw-visible-, and morphological seam-band PSNR/SSIM, full-image and hole-region LPIPS, known-region drift, risk expected calibration error (ECE), area under the risk-coverage curve (AURC), and a projection-defect taxonomy. The final study will additionally report DISTS, depth AbsRel/RMSE, disparity endpoint error, normal error, latency, throughput, peak memory, and parameter count. Results will be stratified by raw overlap, hole ratio, depth validity, instrument presence, and sequence. For both main comparison tables, AlexNet LPIPS receives RGB tensors in $[0, 1]$ and internally maps them to $[-1, 1]$; H-LPIPS averages its spatial map over the same support used by H-PSNR and H-SSIM. This support is the tissue-hole intersection on iMED and the raw-DSS hole itself on EndoVis.

Renderer and reliability. We use $\Xi = (-1, 0, 1)$ and $(a_k) = (0.2741, 0.4518, 0.2741)$; the display percentile in Fig. 3 is $q_{\text{vis}} = 99$ and affects visualization only. Before projection, every sampled depth hypothesis z_{qk} is clipped to $[1, 500]$ mm. Hypotheses affect the radius-one integer neighbourhood, with $\tau_z = 2$ mm and $S_{\min} = 0.03$. Each footprint r_{qk}^α is clipped to $[0.35, 2.5]$ px, with $b = b_n = 0.75$ px. Spatial and final weights no greater than 10^{-5} and 10^{-8} are pruned, and $\epsilon_{\text{dss}} = 10^{-8}$ stabilizes normalization. Coverage uses $C = 1 - \exp(-S)$, and target confidence uses $\kappa = C \exp[-V/(4\tau_z^2) - \mathcal{H}]$. Weight entropy is divided by the logarithm of at least two retained contributions before it is coupled to depth variance.

The reliability encoder fuses full-resolution residual and quarter-resolution context branches to output $d_f = 16$ feature channels. Its fallback median is $s_D = 100$ mm, and valid depth is normalized as $\tilde{D}_s = \text{clip}(D_s/s_D, 0, 5)$ before missing values are set to zero. Measured and completed confidence occupy $[0.5, 1]$ and $[0, 0.35]$, respectively. The measured-depth correction ratio is bounded by 0.1, whereas the completion branch has a 0.05 positive offset. Depth and uncertainty are clipped to $[1, 500]$ mm and $[0.25, 20]$ mm.

Router and experts. The condition maps use normalized variance $\bar{V} = V/(V + \tau_z^2)$ and normalized depth $\tilde{D} = \text{clip}(\bar{D}_t/s_D, 0, 5)/5$. The analytic prior is clipped to $[0, 1]$ and then to $[\epsilon_g, 1 - \epsilon_g]$ before its detached logit ℓ^0 is evaluated. For Eq. (5), the prior weights are $(\alpha_\kappa, \alpha_V, \alpha_H) = (0.65, 0.20, 0.15)$, $\epsilon_g = 10^{-4}$, and $\eta_g = 0.25$; the residual router is zero-initialized. The transport RGB residual is bounded by 0.15. The four-resolution synthesis network uses 3×3 local branches and five output channels; its metric depth is $[\text{softplus}(\cdot) + 0.05]s_D$. The shared base width is 32.

Routing masks and losses. The routing tests use $S_{\text{rel}} = 0.35$, $V_{\text{rel}} = 9 \text{ mm}^2$, and $\mathcal{H}_{\text{rel}} = 1.0$; regional-mask cleanup uses a 7×7 opening kernel. Since collision entropy lies in $[0, 1]$, its configured hard threshold is neutral, although entropy still affects target confidence and soft routing. A separate 7×7 dilation-erosion kernel extracts the training seam. The Charbonnier constant is 10^{-3} , the source-depth

likelihood uses a 10^{-3} scale floor, and target relative-depth error uses a 1-mm denominator floor. Equation (9) uses $(s_{\min}, s_{\max}) = (0.01, 0.50)$ in normalized-RGB units. For the backward cycle, the depth scale is $\text{clip}_{[0.25, 20]}[0.25 + 20 \text{sg}(\mathbf{U}_t)]$ mm and its confidence is $\text{clip}_{[0.05, 1]}(1 - \mathbf{U}_t)$.

Optimization. We trained for 10,000 AdamW steps with learning rate 2×10^{-4} , weight decay 10^{-4} , and three-GPU distributed training. The per-GPU batch size is 1 with four-step gradient accumulation, giving an effective batch size of 12. We use BF16 mixed precision, maximum gradient norm 1, and seed 6666. The implemented loss weights $(\lambda_h, \lambda_v, \lambda_b, \lambda_\nabla, \lambda_d, \lambda_c, \lambda_r, \lambda_u)$ are $(3, 1, 2, 0.2, 0.2, 0.2, 0.1, 0.1)$.

[TODO: Record the final checkpoint, model-selection rule, GPU model, training time, inference warm-up, precision, exact parameter/FLOP counts, and all baseline versions and checkpoints.]

C. Main Comparison

We evaluated every method on the same seven-scene iMED validation split. Each reported method produced one prediction for all 1,392 frames, and every prediction was re-evaluated against the same target image, non-tool tissue support, and canonical raw DSS hole mask from the primary run; the latter covered 23.55% of an image before tissue intersection, while the resulting H support covered 21.74% on average. For the three previously incomplete Gaussian baselines, we completed the missing inference and retained all outputs without frame-level rejection. This includes 398 degenerate EndoGS predictions and 387 degenerate EndoGaussian predictions, so their scores reflect deployment failures rather than a success-only subset. The Gaussian pipelines serve as reference reconstruction baselines rather than matched feed-forward completion models. Endo-4DGS uses the 30,000-step renders for five scenes and the only complete 14,000-step renders for the remaining two scenes. Table I reports the frame-wise mean and sample standard deviation. The DistriNVS row evaluates the primary soft-fusion output directly, without output hard copying or small-component repair. An adapted Hunyuan-DiT checkpoint is included only as an in-domain diagnostic. The final adaptation stage directly used five evaluation sequences, which contain 994 evaluated frames. Its ControlNet initializer was trained on a sixth sequence, while the seventh sequence (199 frames) was used for validation. Hence none of the seven sequences in Table I is held out for this model.

The primary soft-fusion DistriNVS output achieved the best held-out mean in five of the six metrics: PSNR, SSIM, H-PSNR, H-SSIM, and H-LPIPS. Relative to the strongest held-out image-completion baselines, it improved full-image PSNR by 3.47 dB and SSIM by 0.16 over LaMa. Inside the raw DSS holes, it improved PSNR by 5.42 dB and SSIM by 0.11 over LCM-LoRA, and reduced H-LPIPS by 0.008. LaMa achieved the lowest full-image LPIPS, 0.35 versus 0.40 for DistriNVS. The in-domain Hunyuan-DiT diagnostic reached 19.52 dB full-image PSNR and 20.87 dB H-PSNR; these descriptive scores are excluded from held-out

TABLE I

COMPARISON ON THE SEVEN-SCENE iMED VALIDATION SPLIT. EVERY METHOD IS EVALUATED ON THE SAME 1,392 TARGET FRAMES WITHOUT FRAME-LEVEL OUTPUT REJECTION. FULL-IMAGE METRICS USE COMMON NON-TOOL TISSUE SUPPORT; H METRICS USE ITS INTERSECTION WITH THE CANONICAL RAW DSS HOLE MASK FROM THE PRIMARY RUN. VALUES ARE FRAME-WISE MEAN $\pm$ SAMPLE SD; BOLD AND UNDERLINING DENOTE THE BEST AND SECOND-BEST HELD-OUT MEANS, RESPECTIVELY. THE GAUSSIAN RUNS USE DATASET GROUND-TRUTH DEPTH. DISTRI NVS USES DIRECT SOFT FUSION WITHOUT OUTPUT HARD COPYING OR SMALL-COMPONENT REPAIR. LDM USES THE STABLE DIFFUSION v1.5 INPAINTING CHECKPOINT; LCM-LoRA ADDITIONALLY USES AN ENDOSCOPY-DOMAIN LoRA TRAINED ON THE TRAINING SPLIT. THE DAGGER-MARKED HUNYUAN-DiT RESULT IS EXCLUDED FROM HELD-OUT RANKING BECAUSE ITS CHECKPOINT LINEAGE HAS TRAINING EXPOSURE TO SIX EVALUATION SEQUENCES AND VALIDATION EXPOSURE TO THE SEVENTH.

Method	N	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
<i>Gaussian reconstruction—complete coverage</i>							
Deform3DGS [36]	1,392	15.83 $\pm$ 1.87	0.33 $\pm$ 0.06	0.48 $\pm$ 0.10	14.01 $\pm$ 1.81	0.31 $\pm$ 0.07	0.63 $\pm$ 0.06
Endo-4DGS [37]	1,392	16.78 $\pm$ 1.65	0.36 $\pm$ 0.04	0.46 $\pm$ 0.08	15.79 $\pm$ 1.85	0.42 $\pm$ 0.14	0.60 $\pm$ 0.06
Free-SurGS [38]	1,392	17.09 $\pm$ 2.14	0.43 $\pm$ 0.07	0.51 $\pm$ 0.07	17.86 $\pm$ 2.88	0.56 $\pm$ 0.05	0.56 $\pm$ 0.06
SurgicalGS [39]	1,392	17.79 $\pm$ 1.95	0.38 $\pm$ 0.07	0.46 $\pm$ 0.07	17.63 $\pm$ 2.87	0.47 $\pm$ 0.16	0.59 $\pm$ 0.06
PR-ENDO [40]	1,392	15.79 $\pm$ 1.27	0.31 $\pm$ 0.07	0.55 $\pm$ 0.07	11.92 $\pm$ 1.71	0.10 $\pm$ 0.06	0.74 $\pm$ 0.08
EndoGS [41]	1,392	12.52 $\pm$ 6.22	0.38 $\pm$ 0.06	0.66 $\pm$ 0.20	12.52 $\pm$ 6.68	0.49 $\pm$ 0.12	0.74 $\pm$ 0.17
EndoGaussian [42]	1,392	14.38 $\pm$ 4.26	0.37 $\pm$ 0.24	0.77 $\pm$ 0.12	15.86 $\pm$ 3.60	0.48 $\pm$ 0.31	0.69 $\pm$ 0.09
<i>Image completion</i>							
LDM [33]	1,392	17.18 $\pm$ 2.83	0.34 $\pm$ 0.06	0.44 $\pm$ 0.08	14.90 $\pm$ 3.48	0.37 $\pm$ 0.11	0.64 $\pm$ 0.13
LaMa [10]	1,392	19.48 $\pm$ 1.44	0.43 $\pm$ 0.11	0.35 $\pm$ 0.06	21.40 $\pm$ 2.32	0.63 $\pm$ 0.05	0.42 $\pm$ 0.08
LCM-LoRA [34]	1,392	<u>19.47 $\pm$ 1.73</u>	0.42 $\pm$ 0.12	0.39 $\pm$ 0.04	<u>22.53 $\pm$ 2.23</u>	0.70 $\pm$ 0.06	0.42 $\pm$ 0.04
Hunyuan-DiT [†] [35]	1,392	19.52 $\pm$ 1.51	0.42 $\pm$ 0.12	0.38 $\pm$ 0.04	20.87 $\pm$ 2.07	0.64 $\pm$ 0.08	0.49 $\pm$ 0.03
DistriNVS (soft fusion; primary)	1,392	22.95 $\pm$ 2.33	0.59 $\pm$ 0.11	0.40 $\pm$ 0.05	27.95 $\pm$ 2.91	0.80 $\pm$ 0.07	0.41 $\pm$ 0.05

comparison because none of its seven evaluation sequences is held out.

Figure 4 provides the matched qualitative counterpart to Table I.

We next compared the five feed-forward methods on the combined held-out EndoVis Datasets 8 and 9. The frozen test set contains 275 source–target pairs satisfying raw overlap > 0.70 , and every row in Table II evaluates exactly these pairs.

LaMa achieved the best full-image and hole-region PSNR, whereas DistriNVS achieved the best SSIM, LPIPS, H-SSIM, and H-LPIPS. Relative to LaMa, DistriNVS was lower by 0.25 dB PSNR, with 0.03 higher SSIM and 0.09 lower LPIPS; in the raw-DSS holes, it was lower by 0.32 dB H-PSNR, with 0.02 higher H-SSIM and 0.13 lower H-LPIPS. DistriNVS also outperformed LDM, LCM-LoRA, and Hunyuan-DiT across all full-image and hole-region measures.

Figure 5 visualizes the same frozen zero-shot protocol as Table II.

D. Ablation Study

Table III reports the nine completed configurations on the same 1,392 frames from the seven-scene iMED validation split. All models were trained for 10,000 steps with seed 6666 and evaluated without small-component repair. Values in the table are frame-wise mean $\pm$ sample SD, and paired 95% confidence intervals below were obtained by resampling the seven scenes. The primary method is the soft fusion (SF) configuration in Eq. (8). The seven completed component variants retain legacy hard composition (HC) and are there-

fore compared only with the HC reference, rather than treated as removals from the primary model. For compact ablation labels, we use depth completion (DC), distributional depth (DD), anisotropic footprints (AF), physics-guided routing (PGR), RGB-D cycle consistency (RCC), uncertainty loss (UL), and trust-mask cleanup (TMC).

The primary soft-fusion configuration achieved 22.949 dB full-image PSNR, 0.589 SSIM, 27.951 dB hole PSNR, 0.803 hole SSIM, and 24.563 dB seam PSNR. Relative to the legacy hard-copy reference, the paired differences were +2.202 dB for full-image PSNR (95% CI, +1.649 to +2.799) and +2.504 dB for seam PSNR (95% CI, +1.512 to +3.612). AURC decreased from 0.0269 to 0.0206, while reliable-region drift increased from zero to 0.047 (paired 95% CI, +0.042 to +0.052) in normalized RGB units. This quality–fidelity trade-off supports soft fusion as the primary output rule.

Within the legacy hard-copy family, replacing distributional depth with a single deterministic sample produced the clearest consistent degradation: full-image, hole, and seam PSNR decreased by 0.075 dB (95% CI, -0.107 to -0.048), 0.367 dB (95% CI, -0.637 to -0.177), and 0.546 dB (95% CI, -0.801 to -0.311), respectively. Removing the physics router left full-image PSNR nearly unchanged (-0.003 dB; 95% CI, -0.012 to $+0.007$) but reduced seam PSNR by 0.241 dB (95% CI, -0.319 to -0.158). Depth completion and anisotropic footprints showed mixed regional effects: their removal reduced full-image PSNR by 0.069 and 0.095 dB, respectively, but changed hole PSNR by +0.203 dB (95% CI, -0.109 to $+0.525$) and +0.170 dB (95% CI, $+0.056$ to $+0.313$). The deterministic renderer

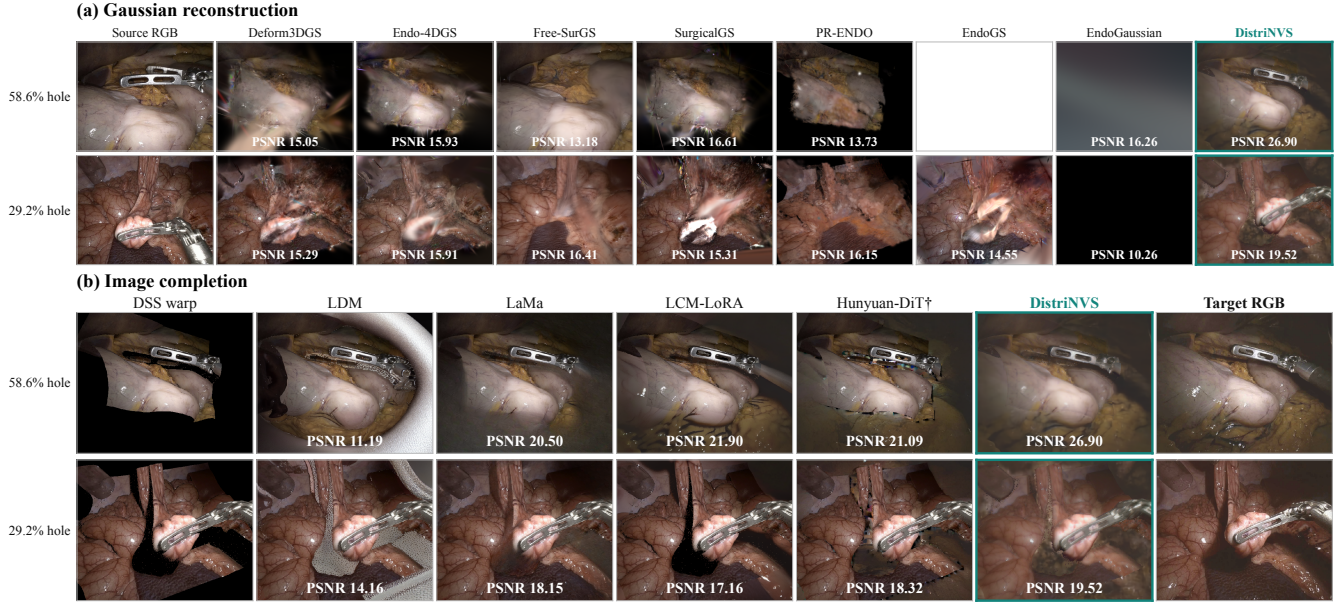


Fig. 4. Qualitative comparison corresponding to Table I. (a) Gaussian reconstruction from source RGB. (b) Image completion from the prepared DSS warp, together with the direct DistriNVS soft-fusion output, for the same two target frames. The rows are the 100th retained pair in frame order from the lexicographically first and last iMED validation scenes; this rule was fixed before any prediction was inspected. Every column within a row uses the same source–target pair. Images are displayed in RGB and resized only to the target dimensions, without cropping or per-image photometric enhancement. Uniform panels in (a) are actual retained baseline predictions rather than missing cells. White labels report each prediction’s per-frame PSNR (dB) under the full-image evaluation protocol of Table I; source, DSS-warp, and target panels are unscored. DistriNVS has the highest displayed PSNR in both rows. The teal border identifies DistriNVS only and does not encode metric rank. Hunyuan-DiT† is the in-domain diagnostic in Table I and is excluded from the held-out ranking.

TABLE II

ZERO-SHOT COMPARISON ON THE COMBINED HELD-OUT ENDOVIS DATASETS 8 AND 9. EVERY METHOD IS EVALUATED ON THE SAME 275 FROZEN PAIRS SATISFYING raw overlap > 0.70 , WITHOUT ENDOVIS ADAPTATION, FRAME REJECTION, OR EVALUATOR-SIDE REPAIR. FULL METRICS USE ALL PIXELS; H METRICS USE THE CANONICAL RAW-DSS HOLE MASK FROM THE PRIMARY RUN. VALUES ARE FRAME-WISE MEAN $\pm$ SAMPLE SD, ROUNDED TO TWO DECIMALS; BOLD AND UNDERLINING DENOTE THE BEST AND SECOND-BEST UNROUNDED MEANS, RESPECTIVELY. LCM-LoRA AND THE DAGGER-MARKED HUNYUAN-DiT† USE SOURCE-DOMAIN iMED ADAPTATION, BUT NEITHER IS ADAPTED ON ENDOVIS. TIME IS SINGLE-FRAME MODEL/PIPELINE LATENCY AT BATCH SIZE ONE ON ONE RTX 4090, MEASURED AFTER WARM-UP ON 10 UNIFORMLY SPACED PAIRS FROM THE SAME FROZEN LIST; MODEL LOADING, IMAGE DECODING, METRIC EVALUATION, AND DISK I/O ARE EXCLUDED.

Method	N	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$	Time (s/frame) $\downarrow$
<i>Image completion</i>								
LDM [33]	275	17.44 $\pm$ 1.86	0.33 $\pm$ 0.14	0.81 $\pm$ 0.17	12.54 $\pm$ 2.44	0.17 $\pm$ 0.05	0.98 $\pm$ 0.11	2.57 $\pm$ 0.15
LaMa [10]	275	23.72 $\pm$ 2.10	0.63 $\pm$ 0.14	0.32 $\pm$ 0.07	21.43 $\pm$ 2.71	0.57 $\pm$ 0.13	0.41 $\pm$ 0.06	0.04 $\pm$ 0.00
LCM-LoRA [34]	275	14.26 $\pm$ 0.94	0.22 $\pm$ 0.04	1.00 $\pm$ 0.08	14.23 $\pm$ 2.61	0.36 $\pm$ 0.13	0.75 $\pm$ 0.07	0.40 $\pm$ 0.04
Hunyuan-DiT† [35]	275	15.79 $\pm$ 2.43	0.31 $\pm$ 0.12	0.85 $\pm$ 0.15	13.24 $\pm$ 3.21	0.36 $\pm$ 0.15	0.75 $\pm$ 0.10	13.04 $\pm$ 0.07
DistriNVS	275	<u>23.47 $\pm$ 2.01</u>	0.66 $\pm$ 0.12	0.23 $\pm$ 0.04	<u>21.11 $\pm$ 2.93</u>	0.59 $\pm$ 0.13	0.28 $\pm$ 0.06	<u>0.08 $\pm$ 0.01</u>

also reduced mean raw overlap from 0.765 to 0.758, so its hole and seam differences include a modest change in the evaluated masks.

Disabling trust-mask cleanup left hole PSNR and hole SSIM effectively unchanged, while increasing seam PSNR by 0.329 dB (95% CI, +0.206 to +0.458) and reducing AURC from 0.0269 to 0.0254. The full-image PSNR change was small and inconclusive (+0.019 dB; 95% CI, −0.005 to +0.044), whereas SSIM decreased by 0.0013 (95% CI, −0.0018 to −0.0008). Cleanup increased the trusted hard-copy ratio from 0.755 to 0.762, but did not improve aggregate seam fidelity under this single-seed evaluation.

Removing the RGB-D cycle caused a small but consistent reduction in full-image PSNR (−0.028 dB; 95% CI, −0.045 to −0.008) and seam PSNR (−0.083 dB; 95% CI, −0.122 to −0.039). Its hole-PSNR change was less conclusive (−0.148 dB; 95% CI, −0.364 to +0.061), indicating that the cycle objective primarily provided a modest regularization benefit. Removing the uncertainty loss reduced hole PSNR by 0.541 dB (95% CI, −0.857 to −0.255) and seam PSNR by 0.102 dB (95% CI, −0.171 to −0.038). More importantly, risk AURC increased from 0.0269 to 0.0365, while risk ECE increased from 0.0236 to 0.9421. Thus, within the legacy hard-copy family, the uncertainty objective was essential for

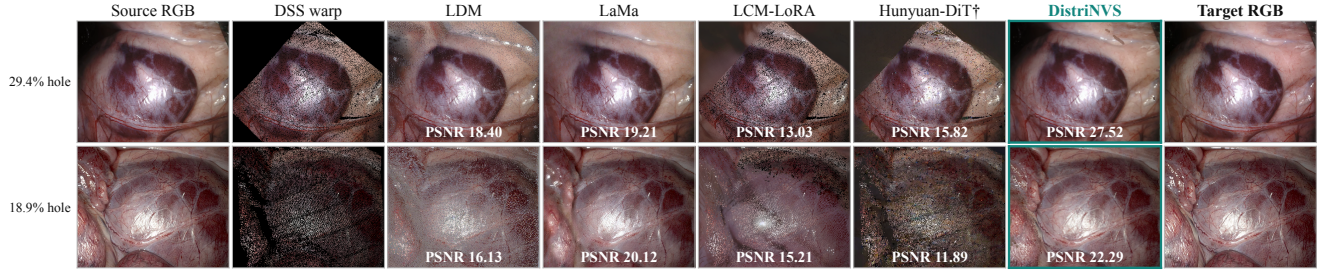


Fig. 5. Qualitative zero-shot comparison corresponding to Table II. For each of EndoVis Datasets 8 and 9, the figure shows the retained pair with the largest positive DistriNVS PSNR margin over the strongest displayed comparator, subject to at least 10% raw-hole area. Completion baselines receive the same prepared DSS warp and binary completion mask, and every column is compared with the same target. White labels report each prediction’s per-frame full-image PSNR (dB); source, DSS-warp, and target panels are unscored. No image is cropped or photometrically enhanced. The teal border identifies DistriNVS only. LCM-LoRA and Hunyuan-DiT† use source-domain iMED adaptation, but no method is adapted on EndoVis.

TABLE III

SINGLE-SEED ABLATION ON THE SEVEN-SCENE iMED VALIDATION SPLIT ($N = 1,392$ FRAMES). H-PSNR AND H-SSIM USE EACH CONFIGURATION’S RAW DSS HOLE MASK; DRIFT IS THE MEAN ABSOLUTE RGB DEVIATION FROM THE DSS WARP ON THE CLEANED RELIABLE REGION. VALUES ARE FRAME-WISE MEAN $\pm$ SAMPLE SD FROM DIRECT OUTPUTS WITHOUT SMALL-COMPONENT REPAIR. BOLD AND UNDERLINING MARK THE BEST AND SECOND-BEST UNROUNDED IMAGE-QUALITY OR RISK MEANS, RESPECTIVELY; DRIFT IS UNRANKED BECAUSE ALL HARD-COPY VARIANTS TIE AT ZERO BY CONSTRUCTION.

Variant	PSNR $\uparrow$	SSIM $\uparrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$	Seam PSNR $\uparrow$	Drift $\downarrow$	AURC $\downarrow$
<i>Primary output rule and legacy reference</i>							
DistriNVS (SF; primary)	<u>22.95 $\pm$ 2.33</u>	<u>0.59 $\pm$ 0.11</u>	<u>27.95 $\pm$ 2.91</u>	<u>0.80 $\pm$ 0.07</u>	<u>24.56 $\pm$ 2.33</u>	0.05 $\pm$ 0.01	<u>0.02 $\pm$ 0.01</u>
DistriNVS (HC; reference)	20.75 $\pm$ 1.90	<u>0.49 $\pm$ 0.13</u>	27.56 $\pm$ 2.78	0.78 $\pm$ 0.07	22.06 $\pm$ 1.70	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
<i>Component variants under legacy hard-copy output</i>							
w/o DC	20.68 $\pm$ 1.89	0.49 $\pm$ 0.13	27.76 $\pm$ 2.48	0.79 $\pm$ 0.06	22.11 $\pm$ 1.66	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o DD ($K = 1$)	20.67 $\pm$ 1.91	0.48 $\pm$ 0.13	27.19 $\pm$ 2.74	0.77 $\pm$ 0.07	21.51 $\pm$ 1.66	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o AF	20.65 $\pm$ 1.92	0.48 $\pm$ 0.13	27.73 $\pm$ 2.71	0.78 $\pm$ 0.06	22.11 $\pm$ 1.73	0.00 $\pm$ 0.00	<u>0.02 $\pm$ 0.01</u>
w/o PGR	20.75 $\pm$ 1.89	0.49 $\pm$ 0.13	27.41 $\pm$ 2.72	0.77 $\pm$ 0.07	21.82 $\pm$ 1.68	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o RCC	20.72 $\pm$ 1.91	0.49 $\pm$ 0.13	27.41 $\pm$ 2.73	0.78 $\pm$ 0.07	21.98 $\pm$ 1.70	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01
w/o UL	20.70 $\pm$ 1.89	0.48 $\pm$ 0.13	27.02 $\pm$ 2.60	0.77 $\pm$ 0.07	21.96 $\pm$ 1.71	0.00 $\pm$ 0.00	0.04 $\pm$ 0.01
w/o TMC	<u>20.77 $\pm$ 1.91</u>	0.49 $\pm$ 0.13	27.56 $\pm$ 2.73	0.78 $\pm$ 0.07	<u>22.39 $\pm$ 1.69</u>	0.00 $\pm$ 0.00	0.03 $\pm$ 0.01

calibrated risk estimates and also improved reconstruction quality in the raw DSS holes.

E. Qualitative, Failure, and Efficiency Analysis

Figures 4 and 5 provide synchronized qualitative counterparts to Tables I and II, respectively.

Efficiency is central to the ICRA framing. Table II shows that DistriNVS is the second-fastest method. Its reported latency includes depth reliability estimation, DSS, routing, both experts, and output composition. Failure analysis will cover specular depth loss, calibration perturbation, thin instruments, large disocclusion, weak texture, and dynamic tissue deformation.

V. DISCUSSION

The ablation separates two complementary design effects. Within the legacy hard-copy family, distributional depth consistently improved full-image, hole, and seam fidelity, whereas the physics router primarily improved seam quality. Depth completion and the anisotropic footprint produced smaller, region-dependent changes. Trust-mask cleanup slightly increased global SSIM and the trusted hard-copy ratio, whereas disabling it improved seam PSNR and AURC.

The output rule yielded the largest difference: the separately trained primary soft-fusion configuration improved full-image and seam PSNR by 2.20 and 2.50 dB and reduced AURC from 0.0269 to 0.0206 relative to the legacy hard-copy reference, at the cost of 0.047 reliable-region drift. This pattern is consistent with hard copying carrying residual transport error to the output, whereas soft fusion permits bounded appearance correction. Because the two output configurations differ during both training and inference, the comparison does not isolate these two effects. The suite also contains one seed, and its component variants were trained under the legacy output rule; their interactions with the primary soft-fusion configuration are therefore not yet established. The fixed EndoVis study compares five methods on 275 common pairs without evaluator-side repair, but remains frame-weighted and unevenly distributed across the ten retained keyframe sequences.

The method does not remove the fundamental ambiguity of unseen tissue. A low risk score indicates model confidence under the training distribution; it does not certify anatomical truth. Moreover, the current formulation assumes known intrinsics and a rigid inter-camera transform for each pair, processes frames independently, and uses an axis-aligned

rather than full-covariance footprint. The combined zero-shot comparison establishes a ranking only under its fixed pair-selection and aggregation protocol; it does not establish uniform generalization across EndoVis sequences, overlap regimes, or anatomy. Large calibration errors, non-rigid motion between nominally synchronized frames, or out-of-distribution anatomy may therefore invalidate the transport statistics. Generated output should remain visibly marked and should not be used as a substitute for a live physical endoscopic view.

VI. CONCLUSION

This paper introduces a reliability-aware formulation of cross-endoscope novel-view synthesis and implements it through distributional surface splatting, reliability-constrained transport/synthesis fusion, and joint appearance-geometry risk modeling. The design turns projection support, depth variance, and collision entropy into operational constraints on what the network may synthesize. No-post validation on the seven-scene iMED split showed that the primary soft-fusion configuration improved full-image and seam PSNR by 2.20 and 2.50 dB over the legacy hard-copy reference. Within the legacy component study, distributional depth provided the most consistent image-fidelity benefit, while physics routing primarily improved transition boundaries. These results support direct reliability-constrained fusion as the primary output formulation. The current evidence is limited to one seed, with component variants evaluated under the legacy hard-copy rule; matched multi-seed runs, sequence-level uncertainty analysis, and broader external datasets remain necessary for stronger comparative claims.

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